Naive Bayes

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Naive Bayes

Independent event

1. coin toss
2. Rolling a Dice

event A → first toss → $\frac{1}{2}$

event B → second toss → $\frac{1}{2}$

H H
H T
T H
T T

$\frac{1}{4}$

$P(A \& B) = P(A) \times P(B)$

$P(A \& B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Dependent events

1. Deck of cards ⇒ 52

event A → Getting a Red card

event B → Getting a Red card

event A ⇒ $\frac{26}{52}$

event B ⇒ $\frac{25}{51}$

(but event B have Diff Prob because of event A)

$P(A \& B) = P(A) \times P(B|A)$

↳ conditional Probability

$P(A \& B) = P(B \& A)$

$P(A) \times P(B|A) = P(B) \times P(A|B)$

$$\boxed{\frac{P(A) \times P(B|A)}{P(B)} = P(A|B)}$$ → Bayes theorem

Independent variable			Dependent variable
Age	Location	Designation	Job
x_1	x_2	x_3	y

$$P(A|B) = \frac{P(A) \times P(B|A)}{P(B)}$$

$$P(y \mid x_1, x_2, x_3) = \frac{P(y) \times P(x_1|y) \times P(x_2|y) \times P(x_3|y) \times P(x_n|y)}{P(x_1) \times P(x_2) \times P(x_3) \times \cdots P(x_n)}$$

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$$\frac{5}{\boxed{6}} = \frac{10}{\boxed{6}}$$

$$P(yes \mid x_1 x_2 x_3) = \frac{P(yes) \times P(x_1 \mid y) \times P(x_2 \mid y) \times P(x_3 \mid y) \cdots P(x_n \mid y)}{\cancel{P(x_1) \times P(x_2) \times P(x_3) \cdots P(x_n)}}$$

$$P(No \mid x_1 x_2 x_3) = \frac{P(No) \times P(x_1 \mid No) \times P(x_2 \mid No) \times P(x_3 \mid No) \cdots P(x_n \mid No)}{\cancel{P(x_1) \times P(x_2) \times P(x_3) \cdots P(x_n)}}$$

All the Probablitis of the target feature will have the same Denominator so you will Remove the Denominator

$$P(y \mid x_1 x_2 x_3) = P(y) \times P(x_1 \mid y) \times P(x_2 \mid y) \times P(x_3 \mid y)$$

$$P(y \mid x_1 x_2 x_3) = \arg\max_{y} \left[P(y) \times \prod_{i=1}^{n} (P x_i \mid y) \right]$$

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